

Joseph Aoun

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I build the automation behind large-scale network infrastructure.

EDUCATION

Columbia University

M.S. in Computer Science, Machine Learning Track

New York, NY

Aug 2026 - Dec 2027 (expected)

- CMA-CGM Excellence Fund Scholarship.

American University of Beirut

B.E. in Computer & Communications Engineering | GPA: 3.8 / 4.0

Beirut, Lebanon

Sep 2021 - May 2025

- Dean's Honor List (2021-2025); MEPI Scholarship; focus on AI and theoretical computer science.

Boston University

Exchange Semester

Boston, MA

Aug 2024 - Dec 2024

TECHNICAL SKILLS

Languages

C/C++23, Python, Java, SQL, Bash

Systems & Tools

Multithreading & concurrency, AWS (CDK, CloudWatch, DynamoDB), CI/CD, Git, Linux, Pytest, JUnit

ML & Data

TensorFlow, TensorFlow Lite, model compression, NumPy, pandas

EXPERIENCE

Amazon Web Services

Software Development Engineer

Dublin, Ireland

Jul 2025 - Aug 2026

- Re-architected a **Tier-1** network-provisioning service in Java from single-threaded execution to a bounded scheduled thread pool with dynamic concurrency control, eliminating head-of-line blocking and parallelizing **100+ requests**.
- Diagnosed and resolved **20+ launch-blocking defects** across race conditions, versioning conflicts, and state filtering, unblocking the production launch.
- Designed end-to-end integration testing from scratch with dedicated hosts and an AWS CDK CI/CD pipeline, executing live provisioning workflows in every commercial AWS region to gate releases.
- Built a per-datacenter circuit breaker deployed across commercial AWS regions, halting dispatch on fault signals to prevent single-datacenter failures from propagating fleet-wide; protects 200+ datacenters in a single region alone.
- Migrated the network-provisioning service to a new internal platform, onboarding **3 upstream services** with **zero downtime**.

Ubilite

Machine Learning Engineer Intern

Beirut, Lebanon

Jun 2024 - Aug 2024

- Compressed a MobileNet face-detection model from **12 MB to 130 KB (92x)** with TensorFlow quantization and pruning while holding **F1 at 85%**.
- Deployed it on Ubilite's Wi-Fi microchip (**< 200 KB SRAM**), enabling on-device inference for low-cost smart-home security devices.

American University of Beirut

Teaching Assistant

Beirut, Lebanon

Jan 2022 - Aug 2024

- Taught and graded **280+ students** across Design & Analysis of Algorithms and Introduction to Programming.
- Co-developed the department's first auto-grading judge, **halving grading time** per problem and accelerating feedback for **500+ students**.

AWARDS & ACHIEVEMENTS

- **ICPC: 23rd** at the 2025 North America Championship, one of 52 qualifying from **1,000+** teams across the U.S. and Canada; **bronze medal** at the North-Eastern North America Regional.
- **IEEEExtreme: 14th** worldwide and **1st** in the U.S. in 2024; top 100 worldwide in 2023 (8,500+ teams each year).
- **Lebanese Collegiate Programming Contest**: two-time silver medalist (80+ teams); qualified twice for the ICPC Arab Collegiate Programming Contest.

SELECTED PROJECT

Chinese Character Transliteration

Python | NLP

- Built a character-level n-gram language model to map English pronunciation to **10,000+ Chinese characters** for keyboard-based text input.
- Improved next-character accuracy from **67% to 87%** using Laplace and Kneser-Ney smoothing.